

Sequential Dependence from Nearest-Neighbor Perturbation: Dressed Embeddings as Bigram-Weighted Averages

Abstract

We perturb the bag-of-words energy model by a nearest-neighbor interaction $\lambda \sum_t \mathbf{e}_{w_t} \cdot \mathbf{e}_{w_{t+1}}$ and work to first order in λ . Absorbing the perturbation into the embeddings defines *dressed* embeddings $\mathbf{e}'_i$ whose shift from the bare values is a bigram-weighted average of all other embeddings: $\delta \mathbf{e}_i = (\lambda/2) \sum_j P_{ij}^{(b)} \tilde{\mathbf{e}}_j$, where $P_{ij}^{(b)}$ is the bigram transition probability determined self-consistently by the model.

Contents

1	The perturbed model	2
1.1	Notation	2
1.2	Unperturbed theory	2
1.3	Nearest-neighbor perturbation	2
2	Bilinear architecture	2
2.1	Embedding and centering	2
2.2	Structure of W	3
3	Polar decomposition and dressed embeddings	3
3.1	Absorbing $\sqrt{W}$ into the embedding	3
3.2	Polar decomposition	3
3.3	Perturbative expansion	3
4	Optimization: finding the best dressed embedding	4
4.1	The reconstruction loss	4
4.2	Grouping by token identity	4
4.3	First-order expansion	4
4.4	Introducing bigram statistics	4
5	Main result: embedding shift as bigram-weighted average	5
5.1	Optimization	5
5.2	The dressed embedding	5
5.3	Centering constraint	5
5.4	Self-consistency: bigram probabilities from the model	5
6	Open questions	5

1 The perturbed model

1.1 Notation

Consider a vocabulary of N tokens and a context window of length L . A sequence is $(w_1, w_2, \dots, w_t, \dots, w_L)$ where t labels the context position. For token type $i \in \{1, \dots, N\}$, define the count fluctuation

$$n_i = \sum_{t=1}^L \delta_{i, w_t} - P_i, \quad (1)$$

where P_i is the expected frequency of token i . The count covariance is $\Sigma_{ij} = \langle n_i n_j \rangle$, and the precision (inverse covariance) matrix is $G = \Sigma^{-1}$.

1.2 Unperturbed theory

The bag-of-words model assigns

$$p_{\text{BoW}}(w_1, \dots, w_L) \propto \exp\left(-\sum_{i,j} G_{ij} n_i n_j\right), \quad (2)$$

with $G_{ij} = (\Sigma^{-1})_{ij}$. The energy is a quadratic form in the count fluctuations, invariant under permutations of positions. The central result of the unperturbed theory is that the optimal embedding vectors satisfy

$$\mathbf{e}_i \cdot \mathbf{e}_j = \Sigma_{ij}, \quad (3)$$

i.e., embedding dot products equal the count covariance.

1.3 Nearest-neighbor perturbation

We add a nearest-neighbor interaction in position space:

$$p(w_1, \dots, w_L) \propto \exp\left(-\sum_{i,j} G_{ij} n_i n_j - \lambda \sum_{t=1}^{L-1} \mathbf{e}_{w_t} \cdot \mathbf{e}_{w_{t+1}}\right) \quad (4)$$

The coupling λ controls the strength of the sequential dependence. We work in the regime $\lambda \ll 1$ with $\|\mathbf{e}_j\| = O(1)$, so the perturbation is small. At $\lambda = 0$ we recover the permutation-invariant BoW model.

2 Bilinear architecture

2.1 Embedding and centering

Expanding the energy (4) in position sums, the full log-probability (up to sequence-independent constants) is $\log p = -\sum_{t,t'} G_{w_t, w_{t'}} - \lambda \sum_t \mathbf{e}_{w_t} \cdot \mathbf{e}_{w_{t+1}} + \text{const}$. This suggests a three-step architecture:

1. **Embed.** Map each token $w_t = i$ to its embedding vector $\mathbf{e}_i \in \mathbb{R}^d$.
2. **Center.** Apply an affine transformation:

$$\tilde{\mathbf{x}}_t = A^{-1}(\mathbf{e}_{w_t} - \bar{\mathbf{e}}), \quad (5)$$

where $\bar{\mathbf{e}} = \sum_j P_j \mathbf{e}_j$ is the mean embedding and A is a (diagonal) scaling matrix.

3. **Bilinear product.** Compute the quadratic form

$$\log p \propto \tilde{\mathbf{x}}^\top W \tilde{\mathbf{x}}, \quad (6)$$

where $\tilde{\mathbf{x}} = (\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_L)^\top$ is the $L \times d$ matrix of centered embeddings and W is an $L \times L$ matrix acting in context space.

2.2 Structure of W

The BoW part contributes an all-to-all coupling (every position pair (t, t') gets weight $G_{w_t, w_{t'}}$). The nearest-neighbor part contributes only to adjacent pairs. The combined weight matrix in context space is

$$W = I + \lambda T, \quad (7)$$

where I represents the BoW contribution (after appropriate normalization) and T is the nearest-neighbor matrix:

$$T_{t,t'} = \delta_{|t-t'|,1}, \quad (8)$$

i.e., T is the adjacency matrix of the 1D chain on L sites.

3 Polar decomposition and dressed embeddings

3.1 Absorbing $\sqrt{W}$ into the embedding

The bilinear form (6) can be rewritten as

$$\tilde{\mathbf{x}}^\top W \tilde{\mathbf{x}} = \tilde{\mathbf{x}}^\top (\sqrt{W})^\top \sqrt{W} \tilde{\mathbf{x}} = \|\sqrt{W} \tilde{\mathbf{x}}\|^2 = \|\tilde{\mathbf{z}}'\|^2, \quad (9)$$

where $\tilde{\mathbf{z}}' = \sqrt{W} \tilde{\mathbf{x}}$. This defines a new set of *dressed* centered embeddings $\tilde{\mathbf{z}}'_t$ obtained by mixing the original centered embeddings across context positions.

3.2 Polar decomposition

Write $W = U_W S_W U_W^\top$ (eigendecomposition, since W is symmetric). Then $\sqrt{W} = U_W S_W^{1/2} U_W^\top$, and the dressed embedding is

$$\tilde{\mathbf{z}}'_t = \sum_{t'} (\sqrt{W})_{t,t'} \tilde{\mathbf{x}}_{t'}. \quad (10)$$

Undoing the centering, this corresponds to a dressed token embedding $\mathbf{e}'_i$ satisfying

$$\boxed{\mathbf{e}'_{w_t} - \bar{\mathbf{e}}' = \sum_{t'} (S_W^{1/2} U_W^\top)_{t,t'} (\mathbf{e}_{w_{t'}} - \bar{\mathbf{e}})} \quad (11)$$

3.3 Perturbative expansion

For $\lambda \ll 1$:

$$\sqrt{W} = (I + \lambda T)^{1/2} \approx I + \frac{1}{2} \lambda T + O(\lambda^2). \quad (12)$$

So the dressed centered embedding is, to first order:

$$\tilde{\mathbf{z}}'_t \approx \tilde{\mathbf{x}}_t + \frac{\lambda}{2} (\tilde{\mathbf{x}}_{t-1} + \tilde{\mathbf{x}}_{t+1}), \quad (13)$$

i.e., each centered embedding receives a correction proportional to the average of its two nearest neighbors in the context window.

Caveat. The mixing (11) acts in **context space** ($L \times L$ matrix on positions), not in token space ($N \times N$ matrix on vocabulary). Since a single token type i can appear at multiple positions, absorbing $\sqrt{W}$ exactly into a position-independent token embedding $\mathbf{e}'_i$ is not possible in general. We therefore seek the *best approximation* via an optimization.

4 Optimization: finding the best dressed embedding

4.1 The reconstruction loss

We want a set of token embeddings $\{\mathbf{e}'_i\}_{i=1}^N$ that best approximates the context-space dressing. Define the loss

$$\mathcal{L} = \sum_{t=1}^L \|\mathbf{e}'_{w_t} - (\sqrt{W} \tilde{\mathbf{x}})_t\|^2. \quad (14)$$

4.2 Grouping by token identity

Let $\gamma(i) = \{t : w_t = i\}$ be the set of positions where token i appears, with $|\gamma(i)| \approx (P_i + n_i) \cdot L$. The loss decomposes as

$$\mathcal{L} = \sum_{i=1}^N \sum_{t \in \gamma(i)} \|\mathbf{e}'_i - (\sqrt{W} \tilde{\mathbf{x}})_t\|^2. \quad (15)$$

4.3 First-order expansion

Using (13), the dressed vector at position t is $\tilde{\mathbf{x}}_t + \frac{\lambda}{2}(\tilde{\mathbf{x}}_{t-1} + \tilde{\mathbf{x}}_{t+1})$. Writing $\mathbf{e}'_i = \mathbf{e}_i + \delta \mathbf{e}_i$, the loss at first order in λ becomes

$$\mathcal{L}(\delta \mathbf{e}_i) = \sum_{i=1}^N \sum_{t \in \gamma(i)} \left\| \delta \mathbf{e}_i - \frac{\lambda}{2} \mathbf{e}_{w_{t+1}} \right\|^2, \quad (16)$$

where we have used the fact that the dominant nearest-neighbor contribution at each position t is $\frac{\lambda}{2} \mathbf{e}_{w_{t+1}}$ (and similarly $\frac{\lambda}{2} \mathbf{e}_{w_{t-1}}$; by symmetry these contribute equally).

4.4 Introducing bigram statistics

For a given token i at position t , the token at position $t+1$ is distributed according to the bigram transition probability $P_{ij}^{(b)} = P(w_{t+1} = j \mid w_t = i)$. Replacing the sum over positions $t \in \gamma(i)$ by an average over bigram statistics:

$$\mathcal{L}(\delta \mathbf{e}_i) = \sum_{i=1}^N (P_i + n_i) \sum_{j=1}^N P_{ij}^{(b)} \left\| \delta \mathbf{e}_i - \frac{\lambda}{2} \tilde{\mathbf{e}}_j \right\|^2, \quad (17)$$

where $\tilde{\mathbf{e}}_j = \mathbf{e}_j - \bar{\mathbf{e}}$ is the centered embedding of token j .

5 Main result: embedding shift as bigram-weighted average

5.1 Optimization

Minimizing (17) with respect to $\delta \mathbf{e}_i$:

$$\frac{\partial \mathcal{L}}{\partial(\delta \mathbf{e}_i)} = 2 \sum_j P_{ij}^{(b)} \left(\delta \mathbf{e}_i - \frac{\lambda}{2} \tilde{\mathbf{e}}_j \right) = 0. \quad (18)$$

Since $\sum_j P_{ij}^{(b)} = 1$, this gives

$$\delta \mathbf{e}_i = \frac{\lambda}{2} \sum_j P_{ij}^{(b)} \tilde{\mathbf{e}}_j. \quad (19)$$

5.2 The dressed embedding

$$\boxed{\mathbf{e}'_i = \mathbf{e}_i + \frac{\lambda}{2} \sum_{j=1}^N P_{ij}^{(b)} (\mathbf{e}_j - \bar{\mathbf{e}}) + O(\lambda^2)} \quad (20)$$

The shift in the embedding of token i is a weighted average of the centered embeddings of all other tokens, with weights given by the bigram transition probabilities $P_{ij}^{(b)} = P(w_{t+1} = j \mid w_t = i)$.

5.3 Centering constraint

The new embedding to first order in λ satisfies

$$\sum_j P_j \delta \mathbf{e}_j = \frac{\lambda}{2} \sum_j P_j \sum_k P_{jk}^{(b)} \tilde{\mathbf{e}}_k, \quad (21)$$

which ensures the dressed embeddings maintain consistent centering.

5.4 Self-consistency: bigram probabilities from the model

The bigram transition probability is determined by the full model (4). Conditioning on $w_t = i$:

$$P(w_{t+1} = j \mid w_t = i) = \frac{P(w_{t+1} = j, w_t = i)}{P(w_t = i)} \approx P_j \cdot e^{-\lambda \mathbf{e}_i \cdot \mathbf{e}_j}, \quad (22)$$

where P_j is the unigram frequency and the exponential factor comes from the nearest-neighbor term. At $\lambda = 0$, $P_{ij}^{(b)} = P_j$ (independence), and the shift (19) reduces to $\delta \mathbf{e}_i = \frac{\lambda}{2} \sum_j P_j \tilde{\mathbf{e}}_j = 0$ (since $\sum_j P_j \tilde{\mathbf{e}}_j = 0$ by definition of centering). The perturbation is self-consistently non-trivial only when $\lambda \neq 0$.

6 Open questions

1. **Higher-order expansion and preferred n -grams.** The present analysis is first-order in λ . At higher orders, the dressing involves trigram and higher n -gram statistics. Can these corrections be organized systematically? Is there a non-perturbative regime where certain n -grams are strongly preferred?

2. **Is the tautological interaction program physical?** The perturbation $V = \lambda \sum_t \mathbf{e}_{w_t} \cdot \mathbf{e}_{w_{t+1}}$ is constructed from the same embeddings that the Gaussian theory determines—it is “tautological” in the sense that the interaction is derived from the unperturbed 2-point function. The self-consistency (22) partially addresses this, but the question remains: does this procedure converge, and does it correspond to a physical (measurable) property of natural language?
3. **Computing bigram statistics from the Gaussian theory.** The result (20) requires the bigram transition probabilities $P_{ij}^{(b)}$, which in principle follow from (22). But this requires knowing $\mathbf{e}_i \cdot \mathbf{e}_j = \Sigma_{ij}$ from the unperturbed theory. The self-consistent loop is: (i) compute Σ_{ij} from the BoW model, (ii) compute $P_{ij}^{(b)} \approx P_j e^{-\lambda \Sigma_{ij}}$, (iii) compute $\delta \mathbf{e}_i$ from (20). Does this iteration converge?